

18. a. H_0 : 3个入口的比例相同

$$P_1 = P_2 = P_3 = \frac{1}{3}$$

H_a :

比例不完全相同

$$\chi^2 = \frac{(83 - 66.67)^2}{66.67} + \frac{(61 - 66.67)^2}{66.67} + \frac{(56 - 66.67)^2}{66.67} = 6.19$$

$$df = 2$$

$$p\text{-value} = P(\chi^2 > 6.17) = 0.0453 < 0.05 \quad \therefore \text{reject } H_0 \neq$$

$$\begin{aligned} b. \quad p^* \pm z_{\frac{\alpha}{2}} \sqrt{\frac{p(1-p)}{n}} &= 0.0415 \pm (1.96) \sqrt{\frac{0.0415(0.9585)}{200}} \\ &= (0.3461, 0.4833) \neq \end{aligned}$$

22. H_0 : Heart attacks occur equally often on each day of the week

H_a : The incidence of heart attacks depends on the day of the week

$$E = \frac{200}{7} = 28.57$$

$$\chi^2 = \sum \frac{(O - E)^2}{E} = 3.63$$

$$df = 6$$

$$p\text{-value} = 0.7266 > 0.05 \rightarrow \text{non reject } H_0$$

18. H_0 : Fatal accident frequency is independent of automobile size

H_a : Fatal accident frequency depends on automobile size

	Small	Medium	Large	
Fatal	61.43	28.04	19.53	$\chi^2 = \sum \frac{(O-E)^2}{E} = 1.886$
Not Fatal	133.51	60.96	42.47	$df = (2-1)(3-1) = 2$
				p-value > 0.05
				$\therefore$ Non-reject H_0

23. H_0 : salary is independent of workdays at home

H_a : salary is dependent of workdays at home

	<1	≥ 1	All at home
<50000	36.21	21.08	10.65
50000-74999	49.01	38.52	14.41
75000-99999	35.2	20.46	19.34
>100000	37.97	22.94	11.59

a. $df = (4-1)(3-1) = 6$

p-value $\propto (\chi^2_{26} > 6.447)$

$= 0.375 > 0.05$

$\therefore$ non reject H_0

b. $df = 6$, $\chi^2 = 6.449$

$0.25 < p < 0.5$ - non reject H_0

$\therefore$ There isn't sufficient evidence that salary depends on telecommuting frequency.